

SET 1 — Smart Agriculture IoT

Scenario

A smart agriculture system is deployed across a large farm to automate irrigation. Each IoT node has a soil moisture sensor, a temperature sensor, and a water flow sensor connected to a microcontroller (Arduino/ESP32). The microcontroller runs firmware that reads sensor data, applies threshold logic, and communicates with a cloud platform using the MQTT protocol over a LoRaWAN gateway.

Soil moisture sensors produce analog voltage signals. The system uses duty-cycled sleep modes to extend battery life. Sensor data is published to specific MQTT topics, and the cloud broker triggers actuators (water pumps) based on subscriptions.

Engineers observe that readings fluctuate even in stable conditions. The LoRa signal weakens at the far end of the farm. Some nodes lose connectivity intermittently, and over-the-air (OTA) firmware updates are planned for all deployed nodes.

Questions

Q1. Which protocol does this system use to send sensor data to the cloud?

- A) MQTT
- B) HTTP REST
- C) FTP
- D) SMTP

Q2. What is the purpose of duty-cycled sleep mode in IoT sensor nodes?

- A) Improve data accuracy
- B) Extend battery life by reducing active time
- C) Increase transmission range
- D) Speed up ADC conversion

Q3. In MQTT, a device that receives messages based on a topic is called a:

- A) Publisher
- B) Broker
- C) Subscriber
- D) Gateway

Q4. Why does the soil moisture sensor output need ADC processing?

- A) To encrypt the signal
- B) To increase signal frequency
- C) To reduce power draw
- D) To convert analog voltage to a digital value the MCU can process

Q5. What is the role of the LoRaWAN gateway in this system?

- A) It bridges LoRa radio packets to an IP network

- B) It stores sensor data locally
- C) It powers the sensor nodes
- D) It runs the MQTT broker

Q6. What causes sensor readings to fluctuate even under stable physical conditions?

- A) MQTT broker overload
- B) ADC quantization noise and sensor-inherent noise
- C) LoRa channel switching
- D) OTA update interference

Q7. Which spreading factor in LoRa increases range but reduces data rate?

- A) SF7
- B) SF1
- C) SF12
- D) SF0

Q8. OTA (Over-the-Air) updates in IoT primarily serve to:

- A) Physically replace hardware components
- B) Increase sensor sampling rate
- C) Reconfigure the MQTT broker
- D) Remotely update device firmware without manual access

Q9. Which IoT architecture layer is responsible for data collection via sensors?

- A) Perception layer
- B) Application layer
- C) Network layer
- D) Business layer

Q10. What happens when an MQTT message is published with QoS level 2?

- A) Message may be lost
- B) Message is delivered exactly once with a 4-step handshake
- C) Message is delivered at most once
- D) Message is broadcast to all brokers

Answer Key

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|-------|--------|
| Q1: A | Q2: B |
| Q3: C | Q4: D |
| Q5: A | Q6: B |
| Q7: C | Q8: D |
| Q9: A | Q10: B |

SET 2 — Smart Home IoT

Scenario

A smart home system integrates motion sensors, smart bulbs, a thermostat, a doorbell camera, and a voice assistant hub. Devices communicate over Zigbee for short-range low-power links and Wi-Fi for high-bandwidth devices like cameras. A local home hub (e.g., Home Assistant) manages automation rules and connects to the cloud.

The thermostat uses a PID control algorithm to maintain target temperatures. The doorbell camera streams H.264 compressed video to cloud storage. All inter-device communication is secured using TLS 1.3. Users access the system remotely via OAuth 2.0 authenticated mobile apps.

The homeowner reports that Zigbee devices near the microwave experience interference. The camera occasionally drops frames, and the hub's local processing struggles when multiple automations trigger simultaneously.

Questions

Q1. Why is Zigbee preferred over Wi-Fi for smart bulbs and sensors in this system?

- A) Zigbee operates on lower power and supports mesh networking
- B) Zigbee has higher bandwidth
- C) Zigbee works only with Apple devices
- D) Zigbee eliminates the need for a hub

Q2. What causes Zigbee interference near a microwave oven?

- A) Magnetic fields from the microwave motor
- B) Both Zigbee (2.4 GHz) and microwaves operate in overlapping frequency bands
- C) Microwave draws too much current
- D) The hub overheats near the microwave

Q3. What does TLS 1.3 provide in this smart home system?

- A) Device firmware compression
- B) Video stream buffering
- C) Encrypted and authenticated communication between devices and cloud
- D) Zigbee channel selection

Q4. OAuth 2.0 in the mobile app is used for:

- A) Video compression
- B) Sensor calibration
- C) Zigbee pairing
- D) Secure delegated access without sharing user credentials

Q5. The thermostat uses a PID algorithm. What does PID stand for?

- A) Proportional-Integral-Derivative

- B) Packet Inspection Device
- C) Protocol Interface Driver
- D) Power Isolation Device

Q6. Why does the camera use H.264 compression for video streaming?

- A) H.264 increases video resolution
- B) H.264 reduces bandwidth and storage requirements while maintaining quality
- C) H.264 is required by Zigbee protocol
- D) H.264 encrypts video automatically

Q7. What is the primary advantage of local processing on a home hub vs. cloud-only processing?

- A) Higher storage capacity
- B) Better video quality
- C) Faster response and functionality even without internet
- D) Automatic firmware updates

Q8. In a Zigbee mesh network, devices that relay messages for others are called:

- A) End devices
- B) Coordinators
- C) Gateways
- D) Routers

Q9. Which layer of the IoT stack handles user-facing applications and dashboards?

- A) Application layer
- B) Perception layer
- C) Network layer
- D) Transport layer

Q10. What is 'edge computing' in the context of this smart home?

- A) Running apps on the cloud only
- B) Processing data locally on the hub rather than sending everything to the cloud
- C) Using Zigbee instead of Wi-Fi
- D) Compressing video before storage

Answer Key

- | | |
|-------|--------|
| Q1: A | Q2: B |
| Q3: C | Q4: D |
| Q5: A | Q6: B |
| Q7: C | Q8: D |
| Q9: A | Q10: B |

SET 3 — Smart Traffic IoT

Scenario

A smart traffic management system is deployed at urban intersections across a city. Inductive loop sensors and LiDAR units detect vehicle presence and speed. Overhead edge computing nodes (NVIDIA Jetson-based) run real-time computer vision models to count vehicles and detect accidents, reducing latency compared to cloud-only processing.

Traffic controllers communicate using the NTCIP (National Transportation Communications for ITS Protocol) standard over a fibre backhaul. Emergency vehicles carry V2I (Vehicle-to-Infrastructure) transponders operating on DSRC (5.9 GHz) to pre-empt signal phases.

The system publishes real-time congestion data via REST APIs consumed by navigation apps. Security engineers are concerned about spoofing attacks on V2I communication. Data retention policies require all incident footage to be stored encrypted for 30 days.

Questions

Q1. Why are edge computing nodes used at intersections instead of sending all data to the cloud?

- A) Edge processing reduces latency for real-time decisions
- B) Cloud storage is too expensive
- C) Cloud cannot process video data
- D) Edge nodes replace inductive loop sensors

Q2. What technology do emergency vehicles use to communicate with traffic signals in this system?

- A) Bluetooth 5.0
- B) DSRC (Dedicated Short-Range Communication) at 5.9 GHz
- C) Zigbee mesh
- D) LoRaWAN

Q3. What is a spoofing attack in the context of V2I communication?

- A) Overloading the traffic controller with requests
- B) Intercepting and reading V2I packets
- C) A fake vehicle/device transmitting false identity or data to infrastructure
- D) Jamming the DSRC frequency band

Q4. Why is LiDAR used alongside inductive loop sensors in this system?

- A) LiDAR is cheaper than loops
- B) LiDAR works only at night
- C) LiDAR replaces fibre backhaul
- D) LiDAR provides 3D spatial data including speed and object classification

Q5. What is the purpose of a REST API in this traffic system?

- A) To expose real-time traffic data to external apps over HTTP
- B) To configure traffic light hardware directly
- C) To encrypt stored footage
- D) To manage V2I transponder pairing

Q6. Encrypting stored incident footage primarily protects against:

- A) Video quality degradation
- B) Unauthorised access to sensitive surveillance data
- C) Network congestion
- D) Edge node overheating

Q7. Which of the following best describes V2I communication?

- A) Vehicle to vehicle data exchange
- B) Video streaming from camera to cloud
- C) Data exchange between a vehicle and roadside infrastructure
- D) Communication between two traffic controllers

Q8. In IoT systems, what does 'latency' refer to?

- A) The amount of data stored per second
- B) The number of devices connected to a network
- C) The encryption strength of a protocol
- D) The delay between data generation and its processing/response

Q9. What is the main advantage of fibre backhaul over cellular for traffic infrastructure?

- A) Fibre provides higher bandwidth and more reliable low-latency connectivity
- B) Fibre is wireless
- C) Fibre is cheaper to install
- D) Fibre supports V2I natively

Q10. Real-time computer vision at the edge node is used to:

- A) Store video permanently
- B) Detect vehicles, count traffic, and identify incidents without cloud round-trips
- C) Manage NTCIP protocol packets
- D) Pair emergency transponders

Answer Key

- Q1: A Q2: B
Q3: C Q4: D
Q5: A Q6: B
Q7: C Q8: D
Q9: A Q10: B

SET 4 — Smart Healthcare IoT

Scenario

A hospital ICU deploys an IoT patient monitoring platform. Wearable patches measure ECG, SpO2, respiratory rate, and skin temperature. These patches use Bluetooth Low Energy (BLE) to stream data to bedside hubs, which aggregate readings and forward them over hospital Wi-Fi to an on-premise edge server.

The edge server runs anomaly detection algorithms and triggers alerts on nurses' tablets via push notifications when parameters exceed clinical thresholds. All data is HL7 FHIR-compliant and integrated with the hospital's Electronic Health Record (EHR) system.

Data is encrypted end-to-end using AES-256. The system must comply with HIPAA regulations. Engineers face challenges with BLE interference in the RF-dense ICU, false-positive alerts from motion artefacts during patient movement, and secure remote access for physicians reviewing data off-site.

Questions

Q1. Why is Bluetooth Low Energy (BLE) used instead of Wi-Fi for wearable patches?

- A) BLE consumes significantly less power, extending wearable battery life
- B) BLE has higher data rate than Wi-Fi
- C) BLE has longer range than Wi-Fi
- D) BLE is required by HIPAA regulations

Q2. What does HL7 FHIR define in the context of this system?

- A) Encryption algorithm for patient data
- B) A standard format for exchanging electronic health information
- C) Wireless protocol for BLE devices
- D) Alert threshold values for ICU monitors

Q3. What regulation primarily governs patient data privacy in this US hospital system?

- A) GDPR
- B) ISO 27001
- C) HIPAA
- D) PCI-DSS

Q4. Motion artefacts in wearable ECG sensors cause false alerts because:

- A) BLE signal is disrupted by movement
- B) The EHR system cannot process movement data
- C) AES-256 encryption adds processing delay
- D) Patient movement introduces noise that mimics abnormal waveforms

Q5. What is the primary purpose of AES-256 encryption in this system?

- A) To ensure patient data is unreadable if intercepted or breached
- B) To compress patient data for faster transmission
- C) To authenticate nurses' tablets
- D) To comply with HL7 FHIR standards

Q6. Why is an on-premise edge server used instead of processing all data in the public cloud?

- A) Cloud platforms cannot handle BLE data
- B) Edge processing reduces latency for critical alerts and keeps sensitive data within the hospital
- C) On-premise servers are cheaper than cloud
- D) Public cloud does not support HL7 FHIR

Q7. SpO2 sensors measure blood oxygen levels using which principle?

- A) Electrical impedance of blood
- B) Ultrasonic pulse reflection
- C) Differential light absorption of oxygenated and deoxygenated haemoglobin
- D) Skin temperature differential

Q8. What causes BLE interference in the ICU environment?

- A) AES-256 encryption overhead
- B) HL7 FHIR data parsing delays
- C) EHR system network traffic
- D) Many BLE devices and other 2.4 GHz equipment operating in a dense RF environment

Q9. Secure remote access for off-site physicians should implement which security measure?

- A) Multi-factor authentication (MFA) and encrypted VPN or zero-trust access
- B) Unencrypted VPN for speed
- C) Shared password across all physicians
- D) Direct BLE connection from home

Q10. In IoT healthcare, what is 'data interoperability'?

- A) Encrypting data before storage
- B) The ability of different systems and devices to exchange and use health data seamlessly
- C) Streaming ECG in real time
- D) Backing up data to multiple servers

Answer Key

Q1: A Q2: B
Q3: C Q4: D

Q5: A	Q6: B
Q7: C	Q8: D
Q9: A	Q10: B

SET 5 — Smart City IoT

Scenario

A metropolitan smart city platform integrates thousands of IoT devices: adaptive LED streetlights with ambient and motion sensors, ultrasonic fill-level sensors in waste bins, distributed air quality monitors measuring CO₂, NO₂, and PM_{2.5}, and smart parking sensors using magnetometers.

All sensor nodes use NB-IoT (Narrowband IoT) as the communication protocol, chosen for deep indoor/urban penetration and low power consumption. Data flows to a centralised City Operating Centre (COC) via a multi-tier IoT platform (device → gateway → message broker → analytics engine → dashboard).

The city's cybersecurity team recently identified that several sensor nodes are running outdated firmware with known CVEs. The analytics platform uses time-series databases and ML models to predict waste bin fill levels and optimise collection routes. Citizens can access real-time parking availability via a public API.

Questions

Q1. Why is NB-IoT chosen over LoRaWAN for this dense urban smart city deployment?

- A) NB-IoT uses licensed cellular spectrum providing better urban penetration and QoS guarantees
- B) NB-IoT is license-free
- C) NB-IoT has higher data rates than LoRaWAN
- D) NB-IoT does not require a SIM card

Q2. What is a CVE in the context of IoT cybersecurity?

- A) A communication protocol for smart sensors
- B) A publicly disclosed vulnerability in software or firmware
- C) A type of encryption standard
- D) A certificate for device authentication

Q3. What type of sensor is used in smart parking to detect vehicle presence?

- A) Ultrasonic fill-level sensor
- B) LiDAR scanner
- C) Magnetometer detecting changes in Earth's magnetic field caused by vehicles
- D) Infrared temperature sensor

Q4. Why does the analytics platform use a time-series database for sensor data?

- A) Time-series databases support 3D spatial queries
- B) Time-series databases encrypt data automatically
- C) Time-series databases replace the message broker
- D) Sensor data is inherently timestamped and time-series DBs are optimised for sequential time-indexed data

Q5. What is the role of the message broker in the IoT platform architecture?

- A) To decouple producers and consumers of data, routing messages between devices and applications
- B) To store time-series data permanently
- C) To run ML prediction models
- D) To manage NB-IoT SIM cards

Q6. PM2.5 in air quality monitoring refers to:

- A) A type of CO2 measurement protocol
- B) Particulate matter with diameter ≤ 2.5 micrometres, posing respiratory health risks
- C) A smart city communication standard
- D) Power management version 2.5 for IoT nodes

Q7. How do ML models improve waste collection in this smart city?

- A) By replacing fill-level sensors with cameras
- B) By encrypting collection schedules
- C) By predicting when bins will be full to enable proactive, route-optimised collection
- D) By reducing the number of bins needed

Q8. What is the main security risk of running outdated firmware on deployed IoT nodes?

- A) Outdated firmware increases power consumption
- B) Outdated firmware reduces sensor accuracy
- C) Outdated firmware causes NB-IoT deregistration
- D) Known vulnerabilities (CVEs) can be exploited by attackers to compromise city infrastructure

Q9. Which of the following best describes a multi-tier IoT architecture?

- A) Layered components (devices, gateways, brokers, analytics, applications) each handling specific functions
- B) A single server handling all IoT functions
- C) Direct device-to-dashboard communication
- D) A mesh network of identical sensor nodes

Q10. What is the benefit of exposing parking availability through a public REST API?

- A) It replaces the need for parking sensors
- B) It allows third-party apps and citizens to access live parking data without accessing backend systems directly
- C) It secures the parking sensor network
- D) It stores parking data in a time-series database

Answer Key

Q1: A	Q2: B
Q3: C	Q4: D
Q5: A	Q6: B
Q7: C	Q8: D
Q9: A	Q10: B